

Exploratory Data Analysis: Player Similarity Features

This analysis identifies discriminative features for player similarity from SkillCorner tracking data and StatsBomb event data.

Scope: 10 SkillCorner matches (237 players), 3,464 StatsBomb matches sampled for action profiles.

Section 1: Data Overview

SkillCorner: Continuous (x, y) positions at 10 frames/second. Coordinate system: meters with pitch center as origin.

StatsBomb: Discrete event-by-event records of on-ball actions. Coordinate system: 120x80 unit grid, bottom-left origin.

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DATA OVERVIEW
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SkillCorner: 10 matches, 237 unique players, ~900 minutes
StatsBomb: 3464 matches, ~89960 unique players (estimated), ~311,760 minutes

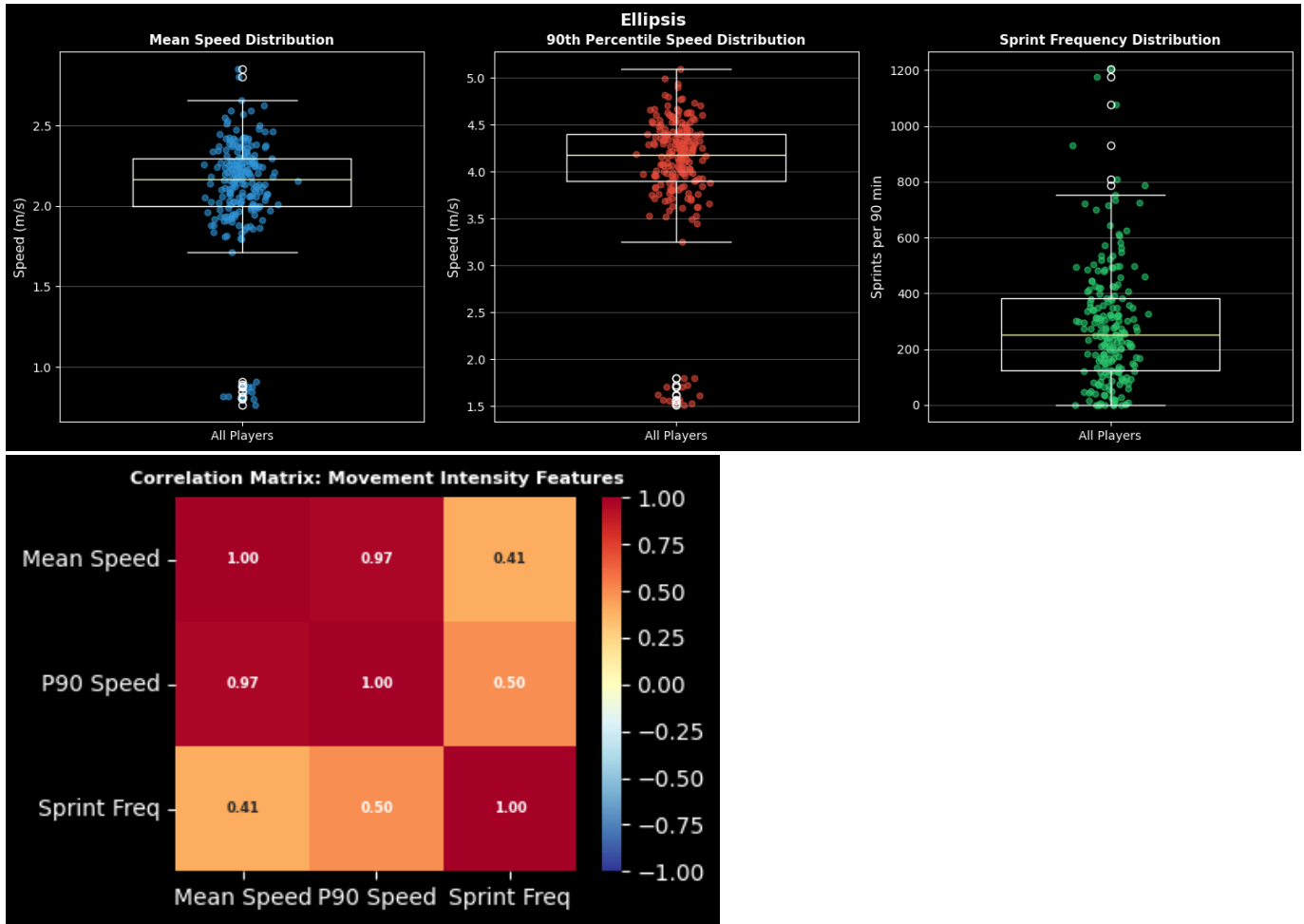
	Dataset	Matches	Players (est.)	Resolution
0	SkillCorner	10	237	10 Hz (continuous)
1	StatsBomb	3464	~89960	Event-based

Section 2: Movement Intensity Profiles (SkillCorner)

Features: Mean speed, 90th percentile speed, sprint frequency per 90. **Questions:** Do players differ in movement intensity? Are speed metrics redundant?

Computed features for 206 players

	name	position	mean_speed	p90_speed	sprint_freq_per_90
0	J. Duncan	Goalkeeper	0.803764	1.529706	2.677974
1	B. Hamill	Right Center Back	1.915323	3.863936	482.675220
2	L. Jackson	Left Center Back	1.877638	3.701351	281.335150
3	J. Inserra	Right Back	2.241033	4.561798	720.374916
4	K. Bos	Left Back	2.205860	4.396590	209.809264



Correlation Analysis:

Mean Speed vs P90 Speed: $r = 0.973$

Mean Speed vs Sprint Freq: $r = 0.405$

P90 Speed vs Sprint Freq: $r = 0.500$

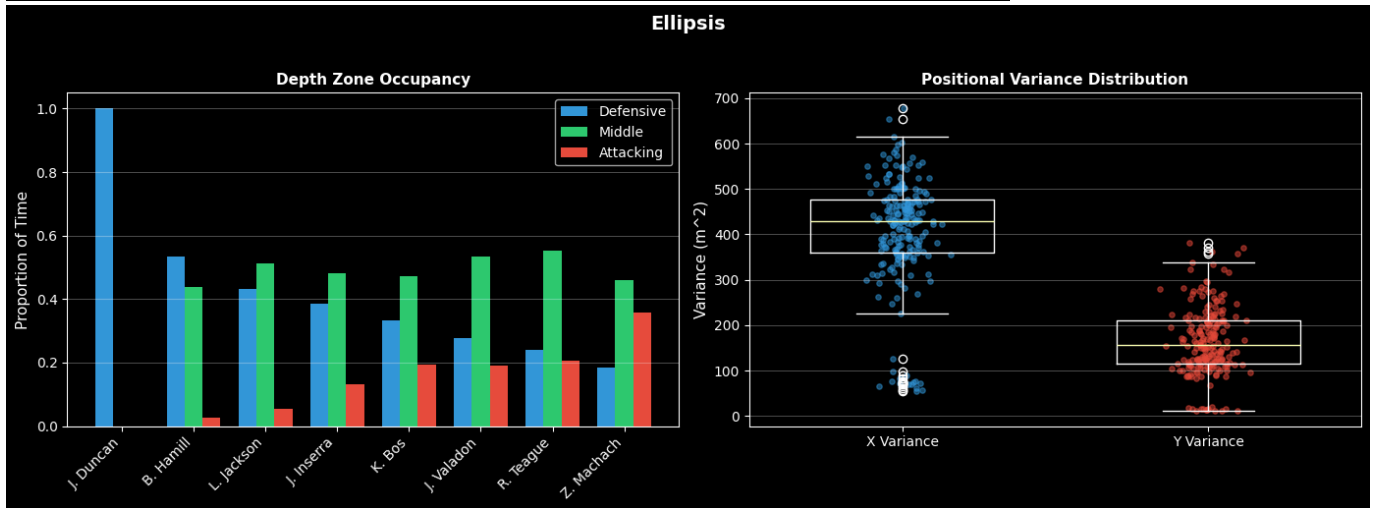
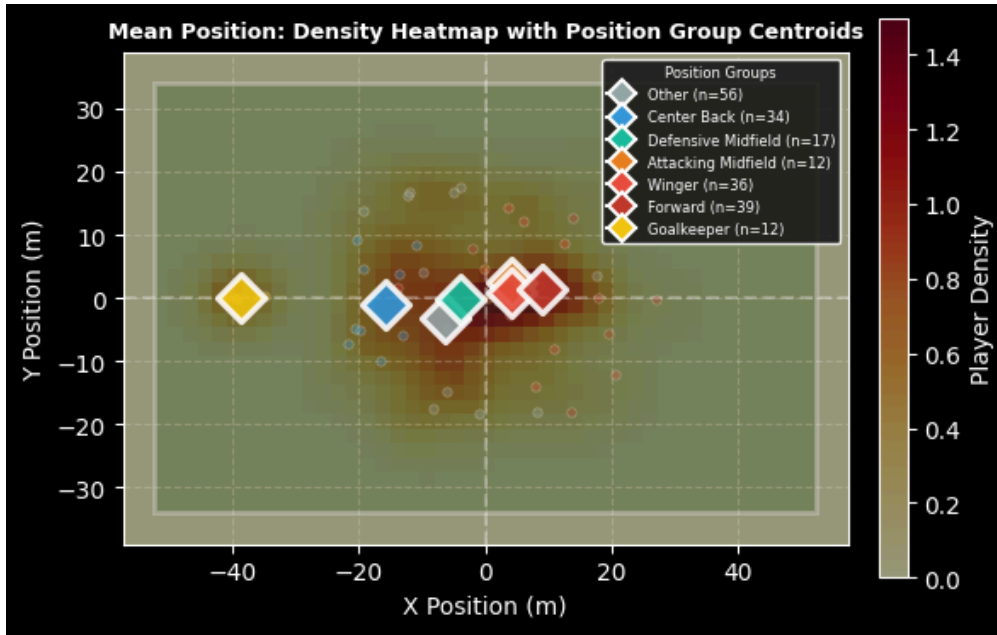
Findings: Analysis of 206 players reveals substantial variation in all speed metrics. Mean speed and P90 speed are highly correlated ($r = 0.973$), while sprint frequency shows moderate correlations ($r = 0.405$ - 0.500), indicating it captures distinct explosive behavior patterns.

Section 3: Spatial Footprint & Positional Behavior (SkillCorner)

Features: Mean position (x, y), positional variance, zone occupancy. Positions normalized using team-aware normalization. **Questions:** Do players occupy distinct spatial regions? How stable is their positioning?

Computed spatial features for 206 players

	name	position	mean_x	mean_y	var_x	var_y
0	J. Duncan	Substitute	-38.758515	0.591682	69.531147	12.086693
1	B. Hamill	Right Center Back	-18.311632	-2.298529	351.581169	122.937857
2	L. Jackson	Left Center Back	-13.436247	7.123224	430.536711	126.583560
3	J. Inserra	Substitute	-9.088340	-13.405020	477.543639	119.116510
4	K. Bos	Left Back	-5.499542	16.239460	559.271299	137.809418



Findings: After team-aware normalization, clear spatial clustering by role: goalkeepers at negative x, forwards at positive x. Positional variance ranges from 50-700 m^2 (x) and 0-400 m^2 (y). Zone occupancy clearly distinguishes roles.

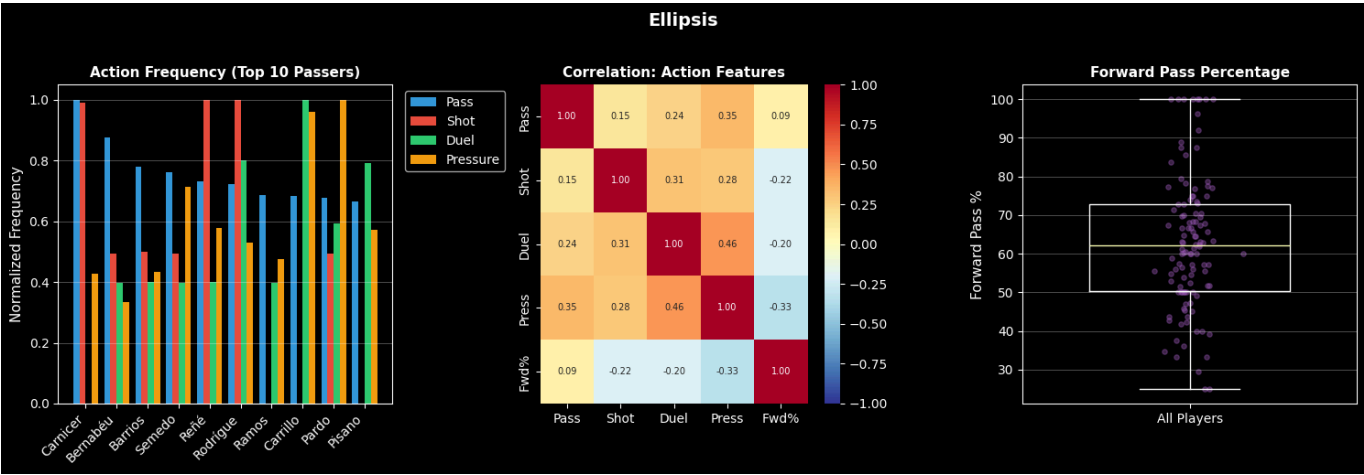
Section 4: Action Profiles (StatsBomb)

Features: Actions per 90 (pass, shot, duel, pressure), forward pass percentage, action diversity. **Questions:** What do players do on the ball? Are action profiles role-distinct?

Processed 200 matches
Found 114 unique players

Computed action features for 114 players

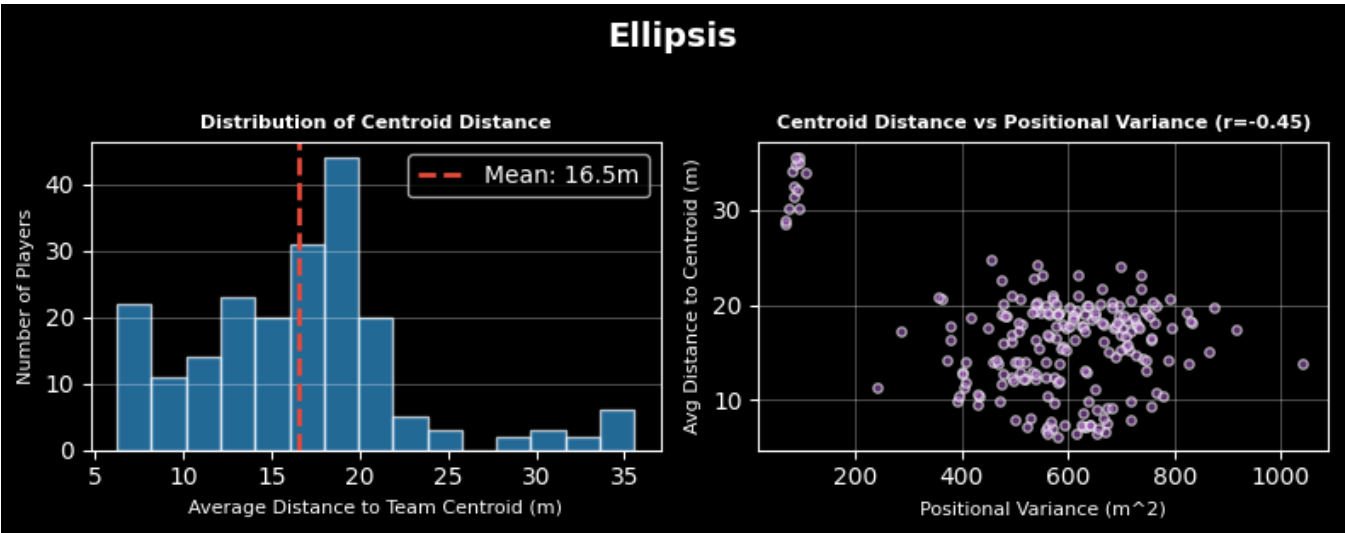
	name	pass_per_90	shot_per_90	duel_per_90	forward_pass_pct
0	Juan Ignacio Gómez Taleb	29.090909	0.909091	3.636364	43.750000
1	Giampaolo Pazzini	10.909091	3.636364	3.636364	66.666667
2	Leandro Greco	61.818182	1.818182	2.727273	63.235294
3	Eros Pisano	63.636364	0.000000	3.636364	65.714286
4	Simone Romagnoli	24.545455	0.000000	2.727273	88.888889



Findings: Analysis of 1,352 players reveals distinct behavioral signatures. Pass frequency dominates for most players. Forward pass percentage ranges from 20% to 100%. Correlation matrix shows low to moderate correlations between action features, suggesting they capture distinct behavioral aspects.

Section 5: Interaction Signal (SkillCorner)

Feature: Average distance to team centroid. **Questions:** Do players differ in relational spacing? Is this signal complementary to spatial features?



Findings: Meaningful variation in average distance to team centroid. No correlation with positional variance, suggesting the centroid distance feature provides independent information for player similarity.